

Hash Collisions, Length Extension & Block-chain Games

The hash threat model

Property	Attacker ideal	Modern answer
Collision resistance	Find any $a \neq b$ with $H(a)=H(b)$	SHA-256: $\sim 2^{128}$ (birthday) – fine
Preimage resistance	Hmm, find x for $H(x)=y$	2^{256} – fine
Second-preimage	Given a , find b same hash	2^{256} – fine
Avalanche	Small change $\rightarrow$ huge change	standard

Birthday bound reality

For n -bit hash, *collisions* appear by $\sim 2^{(n/2)}$ (birthday paradox). MD5 (128-bit) $\rightarrow 2^{64}$ $\rightarrow$ broken. SHA-1 $\rightarrow 2^{80}$ $\rightarrow$ broken in practice (SHAttered, 2017). SHA-256 destined-OK. The message: hash length $\neq$ key length.

Length-extension attack

Merkle–Damgård structure (MD5/SHA1/SHA2) lets you append to a message given $H(m)$ but not m – IF H is used naively as a MAC:

```
MAC = SHA1(secret || message)
attacker learns, without secret, SHA1(secret || message || append)
-> forge a 'valid' signature for a modified message
```

Fix: HMAC (keyed, length-immune), or SHA-3/tree-hash for keyed use.

Block-chain connections

Proof-of-work exploits hash difficulty for consensus (find nonce)

Chaining: each block hashes the previous (mutation detection)

51% + reorg attacks use hash-rate races – availability/extortion vectors

Smart contract bugs deserve their own session (in M40)

In your kit

```
# collision check on legacy files (lab): sha1sum vs sha256sum
# avert: prefer HMAC-SHA256 for MAC; SHA-512 output for signatures
# integrity commands: sha256sum, hmac-calc patterns
```

When an inherited system 'scores a hash' for integrity, ask: is it HMAC? Is the length immune? Is the *output* ≥ 256 -bit? Collision/padded answers win audits.

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